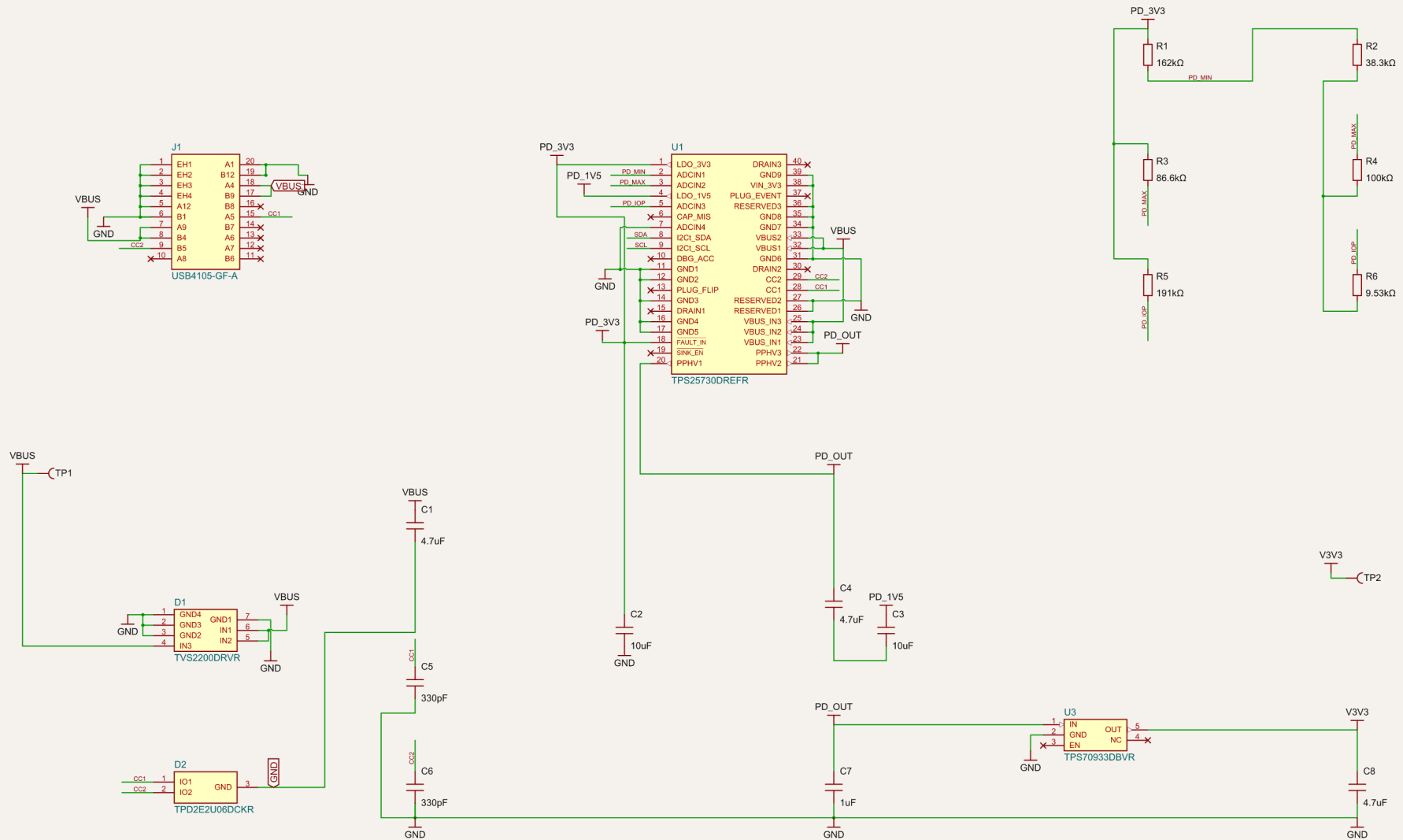


U1: stand-alone PD sink; no NVM. ADC straps 3/5/1/0  
 15 V min/max, 1 A operating / 1.5 A maximum; I2C 0x21  
 PPHV can carry default 5 V: U2 lockout + firmware qualification required.

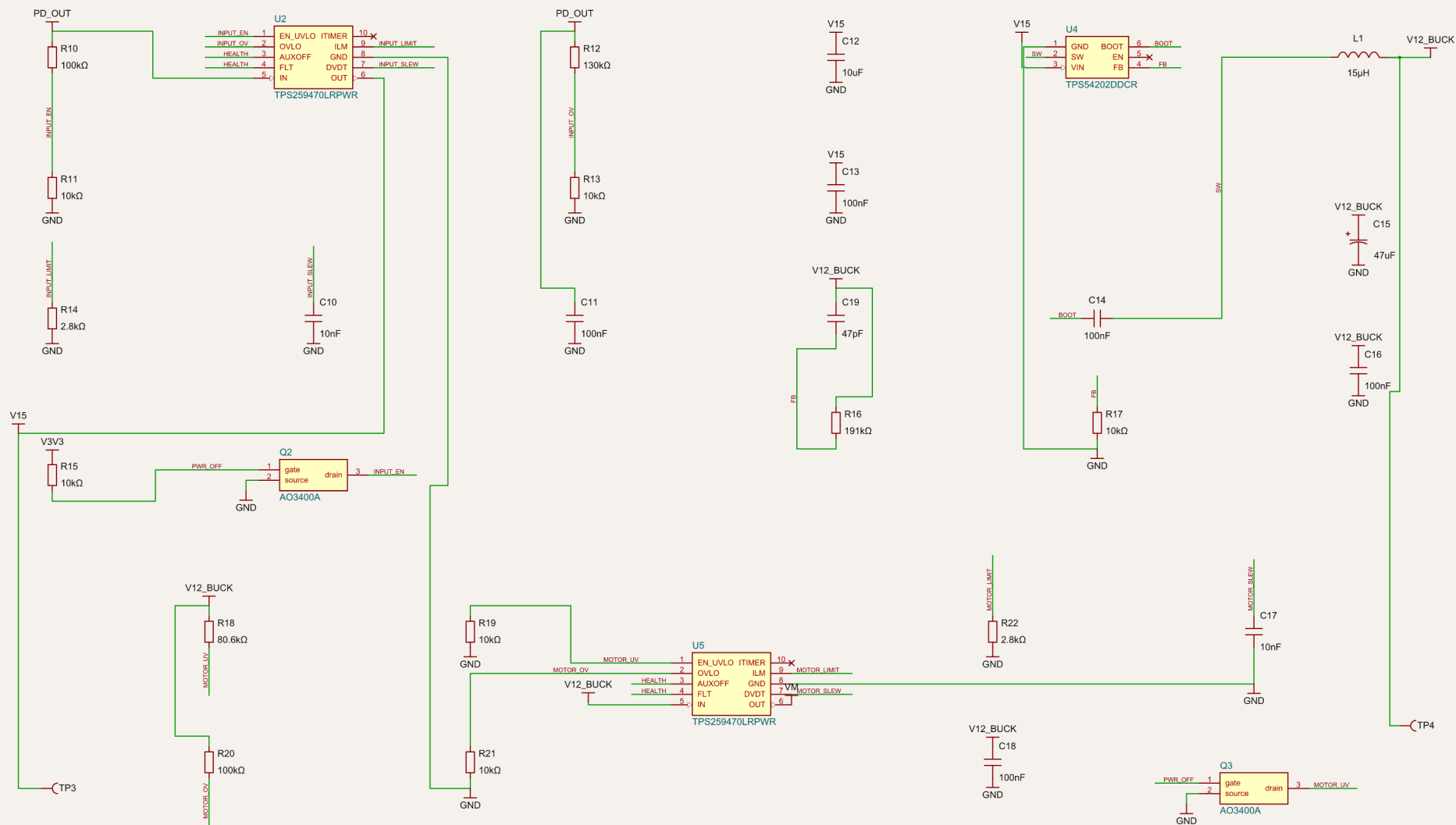


D1: 22 V VBUS flat clamp; surge margin requires qualification.  
 D2: CC-only 5.5 V ESD. Never connect D2 to VBUS.  
 J1 shield tied directly to local ground; USB data unused.

U3: 3.3 V / 150 mA control LDO, 30 V input rating.  
 EN floating = enabled. Intended control load <15 mA.  
 Power present does not mean motor ready.

U2: 2.7–23 V eFuse; 1.19 A nominal limit; latched faults.  
 UV  $\approx 13.2$  V / OV  $\approx 16.8$  V. 0.2 V/ms soft start.  
 Q2 defaults ON to inhibit power until a valid PD contract.

U4: TPS54202 buck, 15 V  $\rightarrow$  11.980 V nominal; 500 kHz.  
 Input 4.5–28 V; IC 2 A rating is not a board rating.  
 15  $\mu$ H / 47  $\mu$ F polymer; FB = 191 k $\Omega$  / 10 k $\Omega$ ; EN internally pulled up.

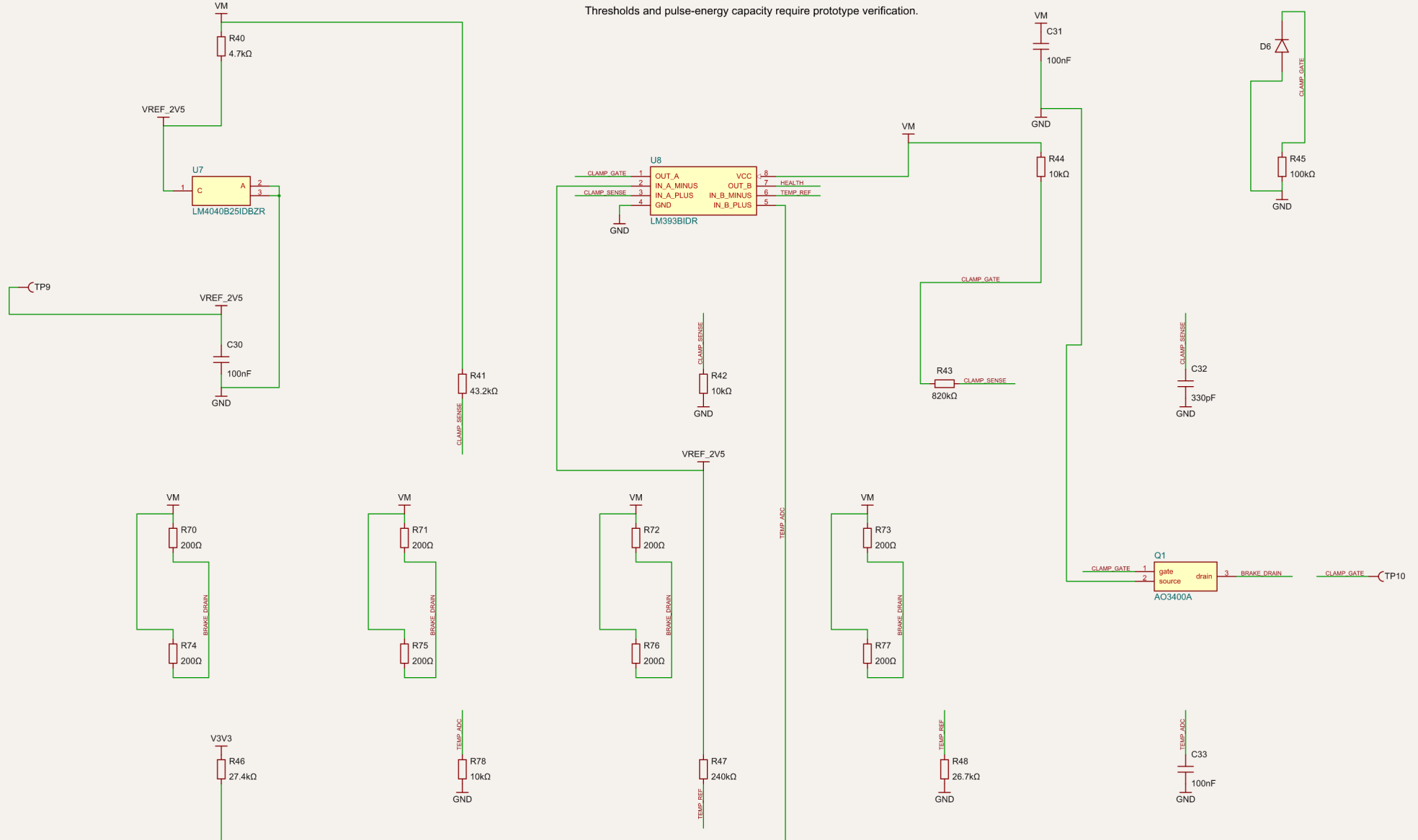


U5: continuous reverse-current blocking isolates regenerated VM from the buck.  
 UV  $\approx 10.872$  V / OV  $\approx 13.2$  V; 1.19 A nominal limit. 470  $\mu$ F bulk is downstream.  
 HEALTH combines open-drain ready/fault signals; U2 disable resets both power stages.

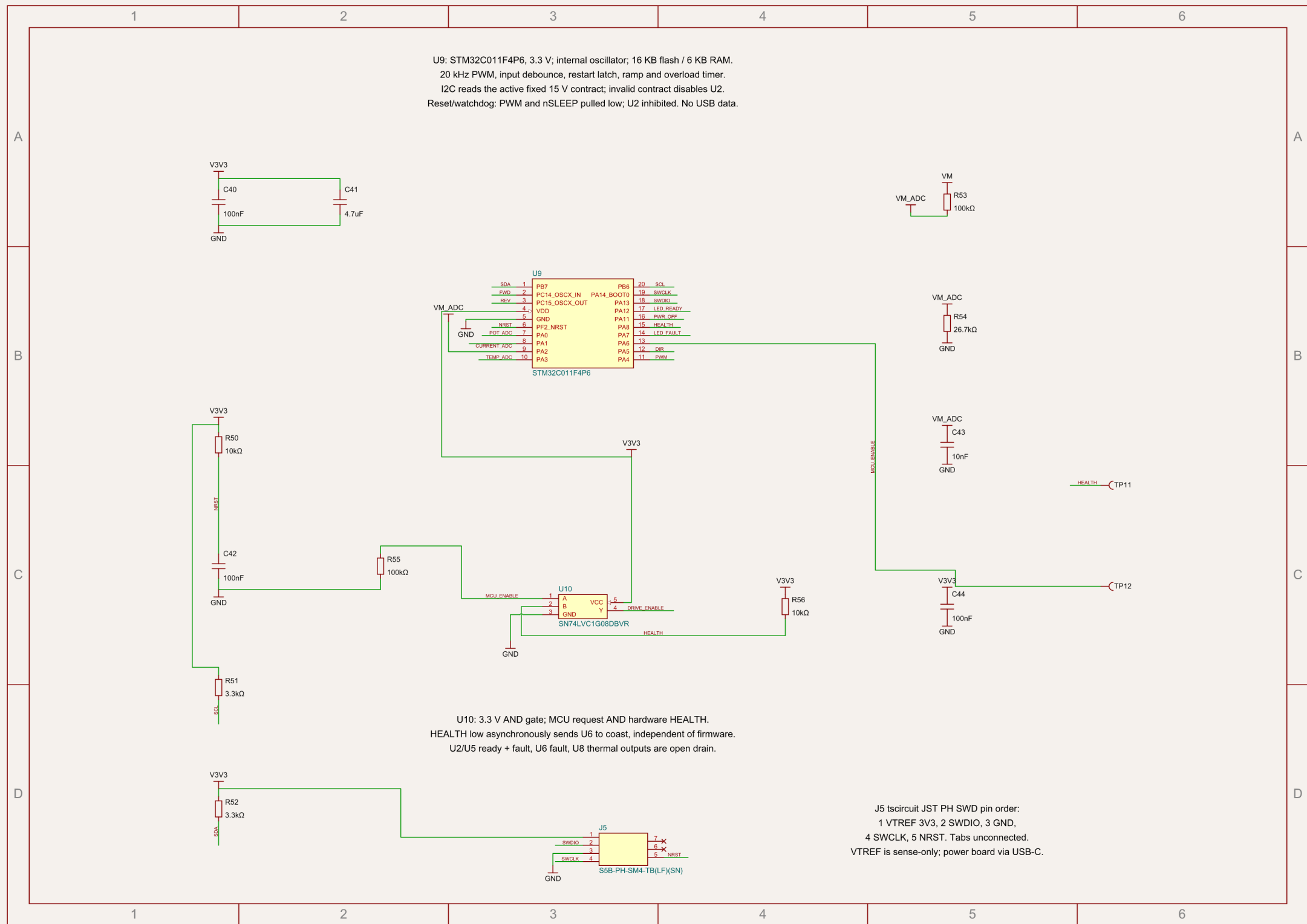


U7: 2.5 V shunt reference, 0.2% initial accuracy.  
 Powered from VM so the clamp works after USB removal.  
 4.7 k $\Omega$  bias  $\approx$  2 mA at 12 V; also sets motor current limit.

U8: VM-powered dual open-collector comparator, 2–36 V.  
 A: hysteretic shunt, nominal ON 13.43 V / OFF  $\approx$  12.95 V.  
 B: NTC overtemperature directly inhibits the driver.  
 Thresholds and pulse-energy capacity require prototype verification.



R70–R77:  $8 \times 200 \Omega / 2 \text{ W}$  in parallel = 25  $\Omega$ ,  $\pm 5\%$ ;  $\approx 7.84 \text{ W}$  total at 14 V.  
 Initial supervised-test envelope:  $\leq 0.5 \text{ J/event}$ ,  $\leq 1 \text{ W}$  average; these are qualification targets.  
 R78: 10 k $\Omega$  NTC B25/50=3380 K. Hardware trip around 70°C; firmware trips earlier.  
 No continuous externally driven load. Safe reversal coast time must be measured.



Panel controls: J3 → 10 kΩ linear pot; J4 → maintained SPDT center-OFF switch.  
 STOP = both contacts open; both closed is a fault. Debounce in firmware.  
 Startup/reconnect requires STOP and knob below 2%, then a deliberate direction selection.  
 Open pot wiper pulls command to zero. Control cables ≤150 mm inside enclosure.

